7	(a)		$\frac{A}{x(A-x)}$	B1 [1]	1.1	Denominator may be multiplied out	

Question			Answer	Marks	AOs		Guidance
7	(b)		**DR** $\int \frac{400}{x(400-x)}\,\mathrm{d}x = \int dt$	**M1**	**1.1a**	Separates variables (inc dx and dt)	400 either side
			$\int \left(\frac{1}{x} + \frac{1}{400-x}\right)\mathrm{d}x = \int dt$	**M1**	**3.1a**	Use of result from **(a)** or method for partial fractions	
			$ln\,x - ln(400-x) = t + c$	**M1** **A1**	**1.1** **1.1**	Attempt to integrate (at least one term correct) All integration correct	
			$ln\left(\frac{x}{400-x}\right) = t + c$			Or $ln\left\|\frac{Ax}{400-x}\right\| = t$	
			When $t = 0$, $x = 100$ so $c = ln\left(\frac{1}{3}\right)$	**M1**	**1.1**	Finding constant (or *A*=3)	
			$t = ln\left(\frac{3x}{400-x}\right)$ When $t = 10$, $10 = ln\left(\frac{3x}{400-x}\right)$	**M1**	**3.4**	Using log laws to combine the c value	
			$\frac{3x}{400-x} = e^{10} \Rightarrow 3x = e^{10}(400-x)$	**M1**	**2.1**	Attempt to rearrange to find *x*	
			400	**A1**	**3.2a**	Must be rounded to nearest whole number	
				[8]			

Question			Answer	Marks	AOs		Guidance
7	(b)		*OR* **DR**				
			$\int_{100}^{X} \frac{400}{x(400-x)}dx = \int_{0}^{10} dt$	**M1**		Separates variables	
			$\int_{100}^{X} \left(\frac{1}{x} + \frac{1}{400-x}\right)dx = \int_{0}^{10} dt$	**M1** **M1**		Use result from (a) or method for partial fractions Attempt to integrate (at least one term correct)	
			$[ln\, x - ln(400-x)]_{100}^{X} = [t]_{0}^{10}$ $\left[ln\frac{x}{(400-x)}\right]_{100}^{X} = [t]_{0}^{10}$	**A1**		All integration correct	
			$ln\frac{X}{(400-X)} - ln\frac{100}{300} = 10 - 0$	**M1**		Limits applied (condone one error)	
			$ln\frac{X}{(400-X)} - ln\frac{1}{3} = 10$			Combining log terms together	
			$ln\frac{3X}{(400-X)} = 10$	**M1**		Attempt to rearrange to find X	
			$\frac{3X}{(400-X)} = e^{10}$	**M1**			
			$3X = e^{10}(400-X)$ $X = 400$	**A1**		Must be rounded to nearest whole number	
				[8]			